A comprehensive analysis of lncRNA prognostic signatures in head and neck squamous cell carcinomas

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1. Introduction

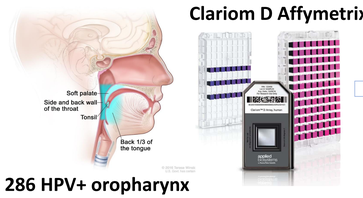
LncRNAs are involved in tumorigenesis and associated with prognosis in many types of cancer. In recent years, many prognostic lncRNA signatures have been developed for survival assessment of head and neck squamous cell carcinomas (HNSC). Most of them are derived from the large cohort available in the TCGA database, without distinguishing HPV-positive from HPV-negative samples.

2. Objective

Since the HPV status determines substantial molecular and clinical differences in HNSC disease, we aimed to test the validity of a set of selected published signatures on our independent cohort composed by 286 samples of HPV-positive oropharynx carcinomas, for which transcriptomic profiles had already been generated with Affymetrix ClariomD microarrays.

3. Methods

ID signature	Journal	# lncRNA	Training
g1	Ann Surg Oncol 2021	3	TCGA (n = 501)
g2	OncoTargets and Therapy 2020	4	GSE65858 (n = 267)
g3	Frontiers in Genetics 2020	6	TCGA (n = 111 LSCC + 11 normal)
g4	Bioscience Reports 2020	15	TCGA
g5	J Cell Physiol 2020	15	TCGA (n = 502)
g6	Bioscience Reports 2018	3	TANRIC (n = 213)
g7	SCieNTiFiC Reports 2018	5	TCGA (n = 250)
g8	J Cell Biochem 2018	4	TCGA
im10	Frontiers in Immunology 2021	9	34 HNSCC cell lines of 19 immune cell types
im11	Cancer Medicine 2021	7	TCGA (n = 297)
im12	Frontiers in Immunology 2021	17	TCGA (n = 227)



286 HPV+ oropharynx carcinomas

Clariom D Affymetrix

Risk score calculation

$$\Sigma\, \alpha_i * X_i$$

α_i = Cox regression coefficient for each lncRNA
X_i = intensity expression of each lncRNA

Raw RNA-sequencing profiles and clinical information of HNSCC patients were downloaded from the TCGA database. Log2 normalized counts with DESeq2 were used for subsequent analyses. In order to test the performance of the risk scores, we applied the concordance index (C-index), using the *survcomp* R package. ROC curve AUCs were calculated with the *pROC* package. Kaplan-Meier curves were built with the *ggsurvplot* function, available within the *survminer* R package.

5. Conclusions

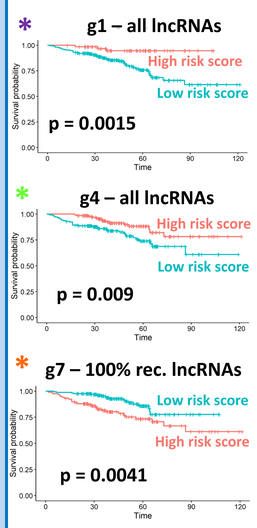
In our large independent cohort of HPV-positive HNSC samples, we were unable to confirm the prognostic validity of most of the selected published lncRNA signatures. The same was partially true when using HPV-positive samples from TCGA. These results point to the need of distinguishing HNSC samples based on their HPV status when developing prognostic signatures.

4. Results

Assessement of the presence of probes for the selected lncRNAs on the ClariomD platform, by means of a Blast search with the 25-mers composing the probes and the putatively recognized lncRNAs sequence (human genome assembly GRCh38).

ID signature	LncRNAs with partial match	LncRNAs with 100% match
g1	LINC02434	AL139327.2, AC126175.1
g2	Z98886.1, AL603962.1	AL645608.1, PIK3CD-AS1
g3	SPRY4-AS1, TTTY14, KLHL7-DT	LINC02154, LINC00528, LNCSRLR
g4	NR1H2, XCR1, NFU1, MIR4435-2HG	KLHDC7B, EMC4, AHDC1, PNPLA4, PEX11A, BRD4, ORC6, AP000229.1, FAM53B, PLEKHG6, ITFG1
g5	MYOSLID, LINC02041, AC012213.4	FOXD2-AS1, WFDC21P, AC073130.1, AL078644.1, LINC01234, AC243773.2, C5orf66-AS1, LINC01305, AC108134.1, ALMS1-IT1, LINC02099, AC019171.1
g6	AC002066.1, AC013652.1	AC016629.3
g7	RP11-474D1.3, RP11-197N18.2	RP11-180M15.7, AC021188.4, RP11-347C18.5
g8	CTD-2506J14.1	RP11-366H4.1, LINC01123, RP11-110I1.14
im10	ENSG00000265148, ENSG00000253230, ENSG00000261888, ENSG00000260244	ENSG00000281358, ENSG00000262089, ENSG00000240889, ENSG00000235304, ENSG00000226806
im11	AC104794.2, AC099343.3, SBDSP1, AC08010.1	AL139158.2, AL031985.3, AL357519.1
im12	AC007878.1, LINC01976, STARD4-AS1, AC103809.1, AC104083.1 (Absent: AC026362.1)	CAVIN2-AS1, LINC00460, AC092115.3, AC068446.2, AL157414.1, LINC01281, GLYCTK-AS1, LINC02325, AL049552.1, AC004461.2, LINC02202

286 HPV+ oropharynx carcinoma



C-index -log(p-value)

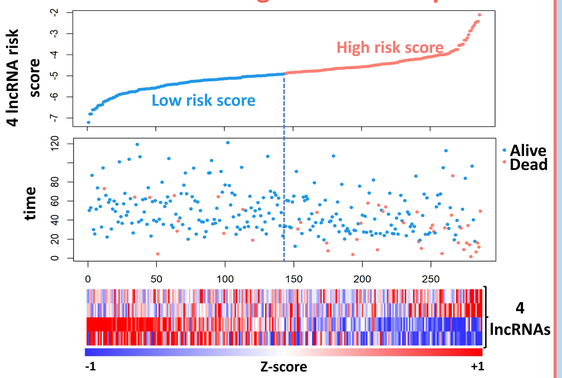
	All lncRNAs	100% recognized lncRNAs
g1	1.45 *	1.53
g2	0.71	0.17
g3	0.06	0.21
g4	2.07 *	2.03
g5	1.02	0.77
g6	0.48	0.11
g7	0.33	3.89 *
g8	0.24	0.04
im10	0.03	0.69
im11	0.15	0.09
im12	0.40	0.43
	2/11	3/11

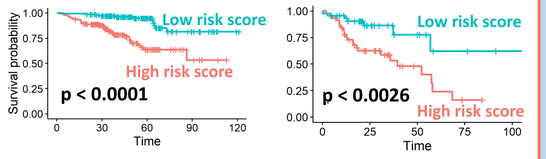
TCGA HNSC

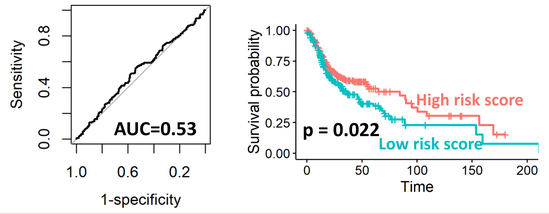
C-index -log(p-value)

	All samples	HPV– samples	HPV+ samples
g1	NA	NA	NA
g2	1.96	1.45	0.82
g3	8.83	6.42	1.69
g4	2.16	1.64	0.71
g5	10.81	5.75	14.73
g6	1.73	0.97	0.98
g7	3.53	2.30	1.70
g8	10.47	5.24	9.53
im10	5.93	3.92	2.85
im11	3.15	4.09	0.02
im12	11.69	5.92	10.12
	10/11	9/11	6/11

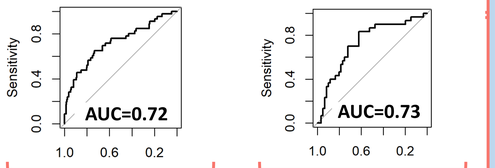
New 4 lncRNA signature development







TEST SET
411 HPV- HNSC TCGA



TRAINING SET
286 HPV+ OSCC

TEST SET
91 HPV+ HNSC TCGA